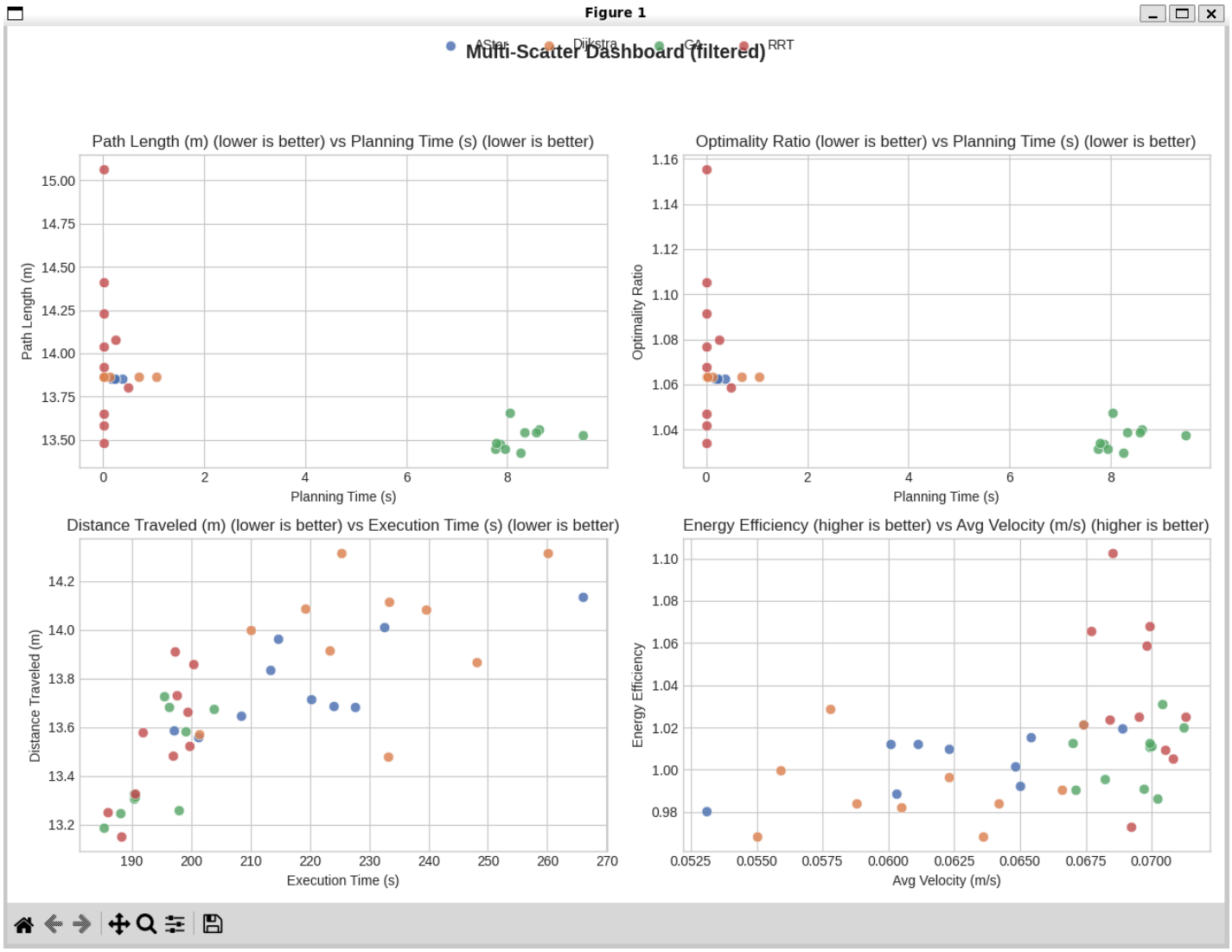
Images from new 10 run per algo,



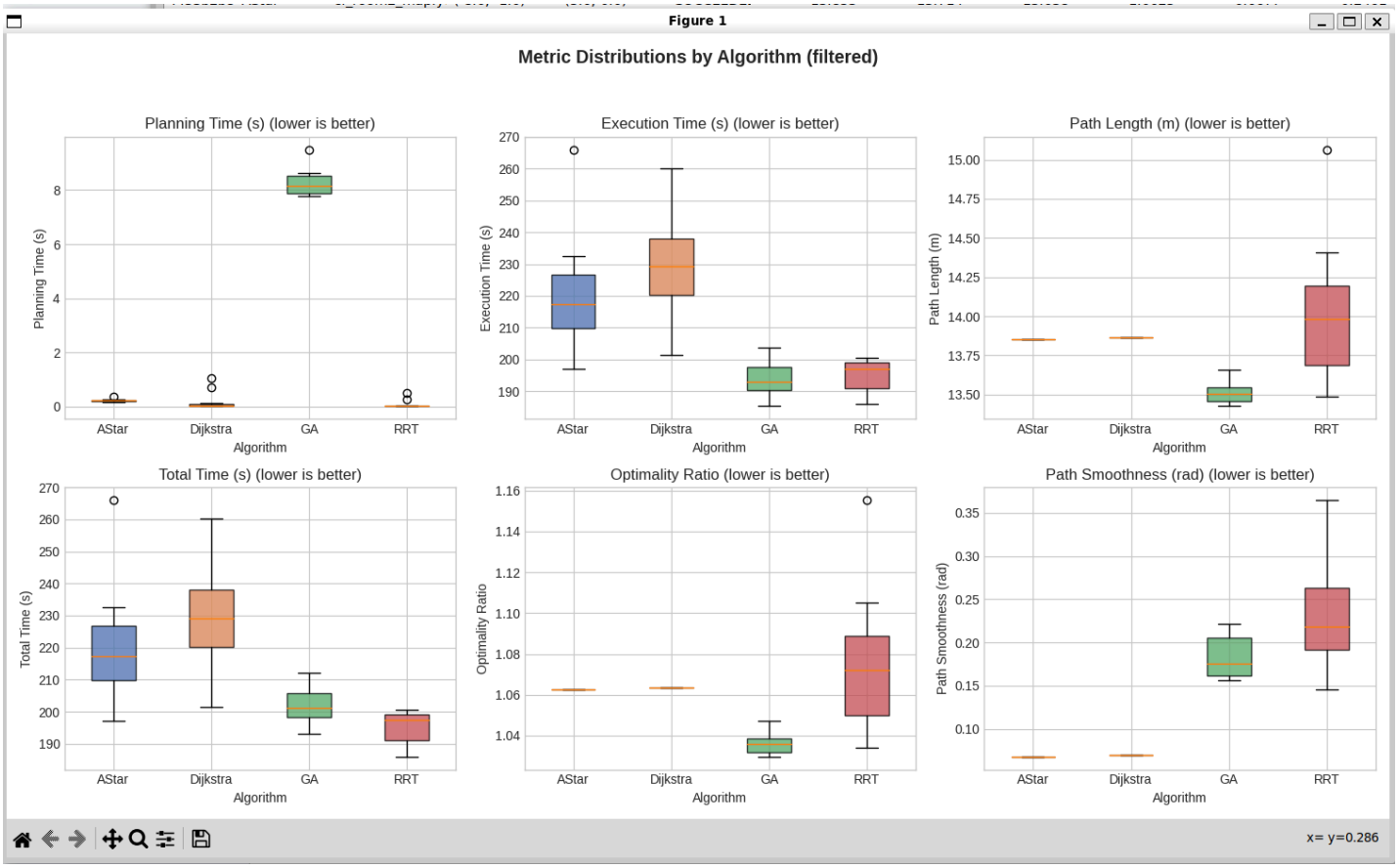


Figure 1
Metric Distributions by Algorithm (filtered)
Planning Time (s) (lower is better)
Planning Time (s)
Execution Time (s) (lower is better)
Execution Time (s)
Path Length (m) (lower is better)
Path Length (m)
Total Time (s) (lower is better)
Total Time (s)
Optimality Ratio (lower is better)
Optimality Ratio
Path Smoothness (rad) (lower is better)
Path Smoothness (rad)
AStar
Dijkstra
GA
RRT
Algorithm
x= y=0.286

Navigation Benchmark Analyzer v3

File Analysis Help

Filters

Algorithm: All Map: All Status: SUCCEEDED Route →: (-8.00,-1.00)→(5.00,0.00) Reset Refresh

40 run(s) [Status=SUCCEEDED, Route=(-8.00,-1.00)→(5.00,0.00)]

Run ID	Algorithm	Map	Start (x,y)	Goal (x,y)	Status	Path Length (m)	Dist Traveled (m)	Straight-Line (m)	Optimality Ratio	Smoothness (rad	Planning Time (s)	Exec Time (s)	Total Time (s)	Completeness	Avg Vel (m/s)	Energy Eff.	Recoveries
0d9aa063	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.646	13.038	1.0625	0.0677	0.2013	208.50	208.51	1	0.065	1.015	0
4453b2b8	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.714	13.038	1.0625	0.0677	0.2461	220.16	220.16	1	0.062	1.010	7
592afd66	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.834	13.038	1.0625	0.0677	0.2187	213.34	213.35	1	0.065	1.001	1
7c9f492b	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.586	13.038	1.0625	0.0677	0.1994	197.06	197.06	1	0.069	1.020	0
9363a975	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	14.012	13.038	1.0625	0.0677	0.2008	232.55	232.56	1	0.060	0.989	10
9e9ac8cd	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.962	13.038	1.0625	0.0677	0.2103	214.67	214.68	1	0.065	0.992	1
b8e07fb6	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.683	13.038	1.0625	0.0677	0.1732	227.56	227.56	1	0.060	1.012	8
d9f547de	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	14.133	13.038	1.0625	0.0677	0.3741	266.00	266.01	1	0.053	0.980	14
e0cce4b5	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.685	13.038	1.0625	0.0677	0.2405	224.02	224.02	1	0.061	1.012	6
ffa25523	AStar	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.853	13.560	13.038	1.0625	0.0677	0.2199	201.21	201.22	1	0.067	1.022	0
04ac87b7	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	14.315	13.038	1.0633	0.0696	0.0139	225.26	225.26	1	0.064	0.968	9
0c7d6777	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	14.084	13.038	1.0633	0.0696	0.0189	239.54	239.55	1	0.059	0.984	9
16f35e61	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	13.865	13.038	1.0633	0.0696	0.0200	248.06	248.07	1	0.056	1.000	6
2bb24307	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	13.914	13.038	1.0633	0.0696	0.0114	223.28	223.28	1	0.062	0.996	5
7ff49516	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	14.114	13.038	1.0633	0.0696	0.0174	233.29	233.30	1	0.060	0.982	6
890da482	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	13.997	13.038	1.0633	0.0696	0.0181	210.11	210.12	1	0.067	0.990	0
8a2d6e7d	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	13.477	13.038	1.0633	0.0696	0.7029	233.13	233.14	1	0.058	1.029	15
b9cf9fcd	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	14.085	13.038	1.0633	0.0696	0.1257	219.27	219.27	1	0.064	0.984	9
bd1e134f	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	13.569	13.038	1.0633	0.0696	1.0416	201.36	201.37	1	0.067	1.022	0
e53ee951	Dijkstra	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.863	14.313	13.038	1.0633	0.0696	0.0176	260.11	260.11	1	0.055	0.969	6
34a455f5	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.543	13.673	13.038	1.0387	0.2114	8.3293	203.77	212.11	1	0.067	0.991	0
40d3b558	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.449	13.306	13.038	1.0315	0.1561	7.7609	190.41	198.18	1	0.070	1.011	0
4d03adbf	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.478	13.327	13.038	1.0337	0.1603	7.8621	190.31	198.18	1	0.070	1.011	0
5ad3e99b	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.656	13.245	13.038	1.0473	0.1657	8.0389	188.06	196.11	1	0.070	1.031	0
a63d63d1	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.561	13.684	13.038	1.0401	0.2210	8.6233	196.27	204.90	1	0.070	0.991	0
b2e3b88b	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.526	13.582	13.038	1.0374	0.1603	9.4951	199.06	208.57	1	0.068	0.996	0
c7030e96	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.425	13.257	13.038	1.0297	0.1837	8.2630	197.87	206.14	1	0.067	1.013	0
cad2902b	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.449	13.185	13.038	1.0315	0.1674	7.9481	185.26	193.21	1	0.071	1.020	0
cb87ef7a	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.543	13.727	13.038	1.0387	0.2114	8.5777	195.42	204.00	1	0.070	0.987	0
f5297dae	GA	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.485	13.314	13.038	1.0342	0.1891	7.7849	190.57	198.36	1	0.070	1.013	0
17d24e6c	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.922	13.150	13.038	1.0678	0.2198	0.0082	188.31	188.33	1	0.070	1.059	0
2a70467f	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.484	13.860	13.038	1.0342	0.2061	0.0075	200.37	200.63	1	0.069	0.973	0
46c215ee	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.650	13.579	13.038	1.0469	0.1453	0.0058	191.91	192.17	1	0.071	1.005	0
7ca4587b	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.585	13.250	13.038	1.0419	0.1795	0.0065	185.96	185.98	1	0.071	1.025	0
9c722383	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	15.064	13.662	13.038	1.1554	0.3650	0.0071	199.37	199.38	1	0.069	1.103	0
abe63e6a	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	14.079	13.732	13.038	1.0798	0.3046	0.2503	197.62	197.87	1	0.070	1.025	0
ac1f6267	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	14.233	13.325	13.038	1.0916	0.2171	0.0064	190.52	190.53	1	0.070	1.068	0
b43d8fa3	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	13.803	13.482	13.038	1.0587	0.1863	0.4913	196.97	197.47	1	0.068	1.024	0
c324b2f1	RRT	cf_room2_map.yaml	(-8.0, -1.0)	(5.0, 0.0)	SUCCEEDED	14.040	13.909	13.038	1.0768	0.2239	0.0095	197.32	197.33	1	0.070	1.009	0

Algorithm Summary – reflects current filter [Route=(-8.00,-1.00)→(5.00,0.00)]

Group by Route

Algorithm	Route	Runs	Path Length (m)	Dist Traveled (m)	Straight-Line (m)	Optimality Ratio	Smoothness (rad)	Planning Time (s)	Exec Time (s)	Total Time (s)	Completeness	Avg Vel (m/s)	Energy Eff.	Recoveries
AStar	(-8.00,-1.00)→(5.00,0.00)	10	13.853 ± 0.000	13.782 ± 0.195	13.038 ± 0.000	1.062 ± 0.000	0.068 ± 0.000	0.228 ± 0.055	220.507 ± 19.518	220.512 ± 19.518	1.000 ± 0.000	0.063 ± 0.005	1.005 ± 0.014	4.700 ± 5.012
Dijkstra	(-8.00,-1.00)→(5.00,0.00)	10	13.863 ± 0.000	13.974 ± 0.279	13.038 ± 0.000	1.063 ± 0.000	0.070 ± 0.000	0.199 ± 0.365	229.341 ± 17.444	229.347 ± 17.445	1.000 ± 0.000	0.061 ± 0.004	0.992 ± 0.020	6.500 ± 4.453
GA	(-8.00,-1.00)→(5.00,0.00)	10	13.512 ± 0.069	13.430 ± 0.211	13.038 ± 0.000	1.036 ± 0.005	0.183 ± 0.024	8.268 ± 0.531	193.701 ± 5.699	201.975 ± 6.049	1.000 ± 0.000	0.069 ± 0.001	1.006 ± 0.015	0.000 ± 0.000
RRT	(-8.00,-1.00)→(5.00,0.00)	10	14.027 ± 0.467	13.547 ± 0.253	13.038 ± 0.000	1.076 ± 0.036	0.232 ± 0.065	0.080 ± 0.164	194.806 ± 5.184	194.942 ± 5.238	1.000 ± 0.000	0.070 ± 0.001	1.036 ± 0.038	0.000 ± 0.000